



S.P.M College, Udantpuri

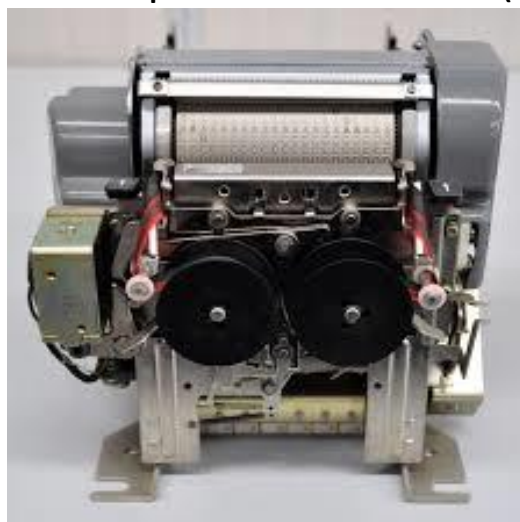
Bachelor Of Computer Application (BCA)

Part -1 (Paper-1)

– Hira Kumar

❖ PRINTER

- The printer is a most commonly used output device.
- It is used to produce the hard copy output.
- It prints characters, symbols & graphics on the paper.
- Printer can be categorised according to the technology used in printer, speed, and approach of printing, colours, language & the quality of printing.
 - ✓ Device Type → Hardware/peripheral/output
 - ✓ Printer Convert → Soft copy to Hard copy
 - ✓ Printer founder/father → Johannes Gutenberg
 - ✓ Speed of printer is measured in → CPM,LPM,PPM
 - CPM → Characters per minute
 - LPM → Lines per minute
 - PPM → Pages per minute
 - ✓ Resolution of Printer = DPI (Dot per inch)
 - ✓ Resolution of image = PPI (Pixels per inch)
 - ✓ Color mode is best for digital work = RGB (Red,Green,Blue)
 - ✓ Color mode is best for printing = CMYK (Cyan,Magenta,Yellow, Black)
 - ✓ First electronic printer → EP-101 (Epson – Japan)



➤ Type of Printers

✓ Impact printer

✓ Non impact printer

a) Impact Printer

- It works on the same mechanism of type-writer.
- It forms a character or image by striking mechanism such as hammer or wheel against to ink ribbon, leaving an image on paper.
- It is oldest technology and still in used.
- It can capable of printing single character or line at the same time e.g. - dot matrix, daisy wheel, chain, drum printer, Line printer.



The Typewriter Database is designed to be printer-friendly, and if you've printed out pages of the site you'll notice that they print out quite a bit differently than they display on-screen. As of last night, I've improved that print-friendliness on the individual gallery pages and the serial number pages.

If you happen to print out the gallery pages for your collection, you'll find that the pages that are printed are optimized for single-page printing, and will render a printout that is suitable for binding into a book or laminating and keeping with your machine during type-ins or other gatherings where you are showing off your machines. The addition of QR codes to the printout to represent the URL links normally associated with your machine to be easily scanned and opened by anyone with a smart phone or tablet that can read QR codes. Right now, the additional photos that go with your gallery aren't printed (who wants the printed thumbnails anyway?), but I'm currently devising a method by which you can print either the single-page info sheet, or alternately print the info sheet and additional pages with the photographs and photo descriptions optimized for printing. Stay tuned on that one. Got so much to do!

➤ CHARACTERISTICS OF IMPACT PRINTER:

- Physical contact with paper to produce output.
- Low cost
- Very noisy
- Very slow in printing
- Low quality print out
- Stand with dusty or extreme environment

b) NON IMPACT PRINTER

- Non impact printer forms characters & images on paper without actually striking the paper.
- The paper and print contact in head/drum come and hence text or image is formed (बनना).

- e.g. –Thermal, Ink jet & laser printer are example of non impact printer.



➤ CHARACTERISTICS OF NON IMPACT PRINTER

- Faster than impact printer
- Ability to change type face automatically.
- High quality output.
- Support transparency.
- More expensive than impact printer.
- Less maintenance than impact printer.

1. Dot matrix printer



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mercial supplier.

- This is a type of Impact Printer. It is a Character printer.
- 1st impact printer.
- Capable to print single character at the same time.
- Forms characters & images as a pattern of dots.

- Contains a print head which moves horizontally across paper.
- Print by **hammering the pins on inked ribbon** to leave ink impressions on the paper.
- Total pin used = 9 / 24 (old = 9, latest = 24)
- Able to print 50 to 500 characters per second.

➤ **ADVANTAGES:**

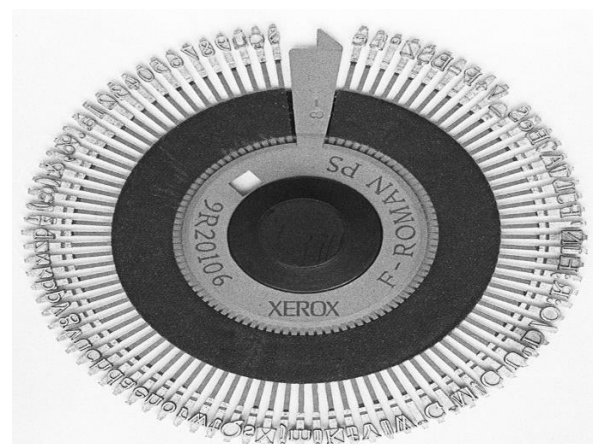
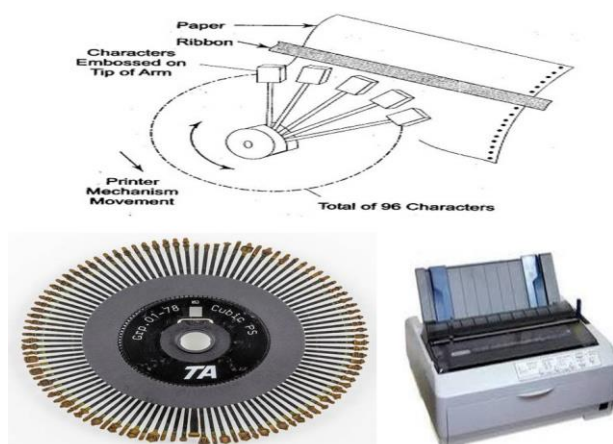
- Low cost & easily available.
- Cheap in cost.
- Can make carbon copy of print out.
- Low maintenance cost.
- Work with any type of environment.

➤ **DISADVANTAGES:**

- Slow in speed.
- Very noisy.
- Cannot work perfectly in graphics.

✓ **USE** :- Bank, Post Office, Railway Ticket....

2. DAISY WHEEL PRINTER:



- This is a type of Impact Printer. It is a Character printer.
- Works same as → Typewriter.
- Able to print a single character at the same time.
- Contain a metal wheel on which the characters & numbers are raised on the each petal(पत्ती) .
- The wheel is rotated very fast when the desired characters arrives at correct position a print hammer strike to produce output.
- Different type of font face can be used by replacing the daisy wheel.
- Able to print bold letter by striking on specific characters twice or thrice.
- Capable to print 10 to 70 characters per second.

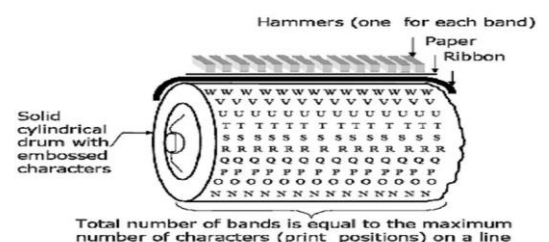
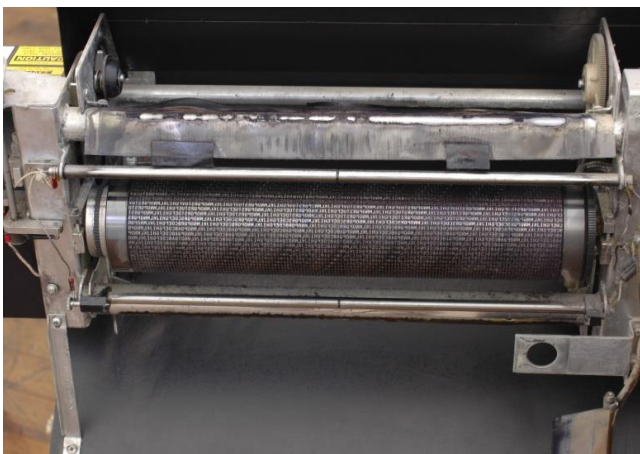
➤ **ADVANTAGES:**

- Low cost.
- Can make carbon copy of print out.
- Low maintenance cost.
- Printing quality is similar to a type writer.
- Able to print bold characters.
- Allows using different font-face in same document.

➤ **DISADVANTAGES:**

- Very slow in speed.
- Very noisy.
- Cannot print graphics.

3. DRUM PRINTER:



- This is a type of Impact Printer. It is a line printer.
- Able to print a line at the same time.
- Consist of a solid cylindrical drum with characters embossed (उभरा हुआ) on it in circular band.
- Each band consists of character set which contains 96 characters.
- Drum rotates fastly when desired characters arrives an appropriate hammer stike on ribbon & character is print on paper.
- Capable to print 300 to 2000 lines per minute.

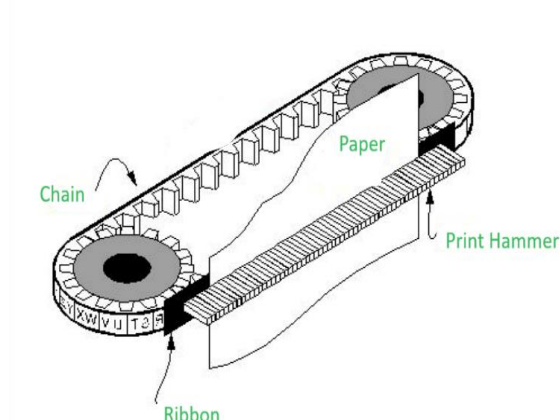
➤ **ADVANTAGES:**

- Low cost than non impact printer.
- Can make carbon copy of print out.
- Low maintenance cost.
- Faster than other impact printer.
- Printing quality is similar to a type writer.

➤ **DISADVANTAGES:**

- Very slow in speed.
- Very noisy.
- Large & heavy.
- Cannot print graphics.
- Only prints predefined set of characters.

4. CHAIN PRINTER:



- This is a type of Impact Printer. It is a line printer.
- Able to print a line at the same time.
- Consist of a metallic chain on which all characters of character set are embossed(उभरा).
- Character set contains 48, 64 or 96 characters.
- Characters are embossed several times.
- Chain rotates at high speed when the desired characters in correct position the hammer strikes & the characters are print on paper.
- Capable to print 400 to 2500 lines per minutes.

➤ **ADVANTAGES:**

- Low cost than non impact printer.
- Can make carbon copy of print out.
- Chain can be easily changed.
- Allowed to print different type font.
- Printing quality is similar to a type writer.

➤ **DISADVANTAGES:**

- Slower than non impact printer.
- Very noisy.
- Large & heavy.
- Cannot print graphics.
- Only prints predefined set of characters.

5. INK-JET PRINTER:



- It's non-impact printer. It's a character printer.
- Forms characters and all kinds of images by spraying drops of ink on to the paper.
- Print head contains 64 tiny nozzles.
- To print a character the printer selectively heats the appropriate set of nozzle as the print head moves horizontally.
- Inkjet printer can either colour or monochrome/Black & White.
- Capable to print 30 to 400 characters per minutes.
- Update version of inkjet is Inktank.

➤ **ADVANTAGES:**

- High quality output.
- Silent during the operation.
- Able to print graphics.
- Able to print any characters & graphics.
- Able to generate colour & monochrome output.

➤ **DISADVANTAGES:**

- Slower than laser printer.
- Cannot make carbon copy of print out.
- Expensive than impact printer.

6. LASER PRINTER:



- It's non-impact printer. It's a page printer.
- Three main components laser beam, a multi-sided mirror, a photoconductive drum & toner.
- To print page laser beam is focused on drum by spinning multisided mirror.
- Drum is electric charged.
- Toner which is composed of oppositely charged ink particles, stick to the drum.
- Then toner focused on the paper with heat & pressure to generate output.
- Low speed laser printer can print 15 to 40 page per minute while high speed laser printer Capable to print 300 to 500 pages per minutes.

➤ **ADVANTAGES:**

- High quality output.
- Very faster in speed.
- Silent during the operation.
- Able to print graphics.
- Able to print any characters & graphics.
- Able to generate colour & monochrome output.

➤ **DISADVANTAGES:**

- Very expensive.
- Cannot make carbon copy of print out.

7. Thermal Printer

- It uses heat to print letters on paper, No any ink /toner use.
- Thermal printer used → thermos chromic paper.
- Founder → Jack kilby in 1965.
- The ink is melted with the help of heat and released on the paper, which prints letters or pictures.
- Old fax machine, POS , ATM, Mall,.....is also a type of thermal printer.
- It is slower (80mm/sec) and more expensive than other printers.
- It is special type of paper is required to use it which is Chemically Treated Paper(ctp).



THE DIFFERENCE BETWEEN IMPACT & NON-IMPACT PRINTER.

Impact printer	Non impact printer
Printing character by striking hammer against ink ribbon to produce output	Printing characters or graphics by spraying ink on paper.
Slow in speed	Faster than impact printer
Work with any environment	Can't work with all environment
Less expensive than non impact printer	More expensive than impact printer
Noisy during printing	Silent during printing
Able to produced carbon copy output	Cant able to produced carbon copy output
e.g.: daisy wheel, drum, chain, dot matrix	e.g.: inkjet, laser

=====HIRA KUMAR=====